

Amazon Order Performance Dashboard

Insights & Strategy Report

1. Project Overview

The objective of this project was to analyze Amazon order data and identify meaningful business insights related to sales performance, customer purchasing behaviour, geographic trends, fulfilment efficiency, and revenue generation.

The analysis was performed using Python for data cleaning and exploratory data analysis, SQL for querying and aggregation, Excel for pivot-based exploration, and Power BI for dashboard visualization.

The dataset contains approximately 125,000 Amazon order records, including information related to order status, product category, size, revenue, fulfilment method, customer location, and promotional activities.

2. Data Preparation Summary

The dataset was cleaned and prepared before analysis. The following preprocessing steps were performed:

- Missing values were identified and handled.
- Duplicate records were removed.
- Date columns were converted to appropriate formats.
- Categorical values were standardized.
- Additional derived columns such as Month and Year were created for trend analysis.
- Identified and treated revenue outliers using the IQR method.

These steps improved data quality and ensured accurate analysis.

The final cleaned dataset was used for SQL analysis, Excel reporting, and Power BI dashboard development.

Dataset Overview

- Total Orders: 125,000
- Total Revenue: ₹77.40 Million
- Average Order Value: ₹620.26
- Product Categories: 9
- States Covered: 47
- Product Sizes: 11

The dataset provides a comprehensive view of customer orders, product performance, shipping operations, and revenue generation.

3. Sales Performance Insights

3.1 Sales Performance Analysis

The Kurta category generated the highest order volume, indicating strong customer demand. However, revenue analysis showed that the Set category contributed the highest overall revenue, followed by Kurta and Western Dress categories. This indicates that while Kurta products sell more frequently, Set products generate higher revenue per order.

These categories represent the primary revenue drivers of the business, suggesting that customer demand is strongly focused on specific product groups.

The most popular product sizes accounted for the majority of orders, indicating strong customer preference patterns. Maintaining inventory for Medium and L sizes is critical to avoid stock shortages.

3.2 Revenue Analysis

The total revenue generated from all orders was 77.40 Million.

Monthly order analysis revealed a rapid increase in demand during April, which recorded the highest order volume. Although order volume remained strong during May and June, a slight decline was observed after April.

This pattern suggests possible seasonal demand fluctuations and highlights the importance of inventory planning during peak periods.

3.3 Geographic Analysis

The state contributing the highest order volume and revenue was Maharashtra, Karnataka, and Tamil Nadu. These states represent the strongest customer markets and should be prioritized for marketing investments.

The city generating the highest order volume and revenue was **Bengaluru**, Hyderabad, and Mumbai. These metropolitan areas demonstrate strong purchasing power and customer engagement.

A large percentage of total sales originated from a small number of metropolitan regions, indicating strong market concentration.

3.4 Order Status Analysis

Most orders were successfully delivered or shipped.

The overall cancellation rate remained relatively low; however, specific categories experienced higher cancellation percentages. While overall performance is strong, the cancellation percentage remains significant and requires attention.

Reducing cancellation rates could improve revenue realization and customer retention.

3.5 Fulfilment Analysis

The analysis revealed that Amazon Fulfilment handled approximately 69.52% of all orders, while Merchant Fulfilment accounted for 30.48%. This indicates that customers rely heavily on Amazon's fulfilment infrastructure.

Amazon Fulfilment contributes significantly to operational efficiency, faster delivery times, and improved customer satisfaction.

3.6 Promotion Analysis

Orders associated with promotional campaigns generally achieved higher average order values compared to non-promotional orders. This demonstrates that targeted promotions positively influence customer spending behaviour.

4. Business Challenges Identified

The analysis identified several important challenges:

1. Revenue concentration within a small number of categories.
2. Higher cancellation rates in specific products.
3. Dependence on a limited number of geographic markets.
4. Performance differences between fulfilment channels.
5. Underperforming categories contributing minimal revenue.
6. Declining order volume after peak demand periods.

Addressing these challenges can improve long-term business growth and operational efficiency.

5. Strategic Recommendations

Strategy 1: Increase Focus on High-Revenue Categories

Allocate additional inventory, marketing budget, and promotional efforts to categories namely Set, Kurta, Western Dress, and Top that consistently generate strong revenue.

Strategy 2: Expand Presence in High-Performing Regions

Increase advertising and customer engagement campaigns in top-performing states and cities. This approach can maximize return on marketing investment.

Strategy 3: Reduce Product Cancellations

Investigate the root causes of cancellations and implement:

- Better stock management
- Improved product descriptions
- Faster shipping processes
- Enhanced quality assurance procedures

Strategy 4: Optimize Fulfilment Operations

Continue prioritizing Amazon Fulfilment due to its strong operational performance and customer trust.

Strategy 5: Improve Promotional Campaign Effectiveness

Use historical promotion performance to design future campaigns. Focus on promotions that increase average order value instead of offering broad discounts.

Strategy 6: Develop Low-Performing Markets

Develop targeted campaigns for lower-performing states and cities to reduce dependence on a limited number of regions.

Strategy 7: Seasonal Inventory Planning

Prepare additional inventory before high-demand periods such as April to prevent stock shortages and lost sales opportunities.

6. Recommendations for New Amazon Sellers

Based on the analysis, new Amazon sellers should:

- Focus on high-demand categories such as Kurta and Set.
- Prioritize high-performing metropolitan markets.
- Maintain inventory for popular product sizes.
- Utilize Amazon Fulfillment services.
- Monitor cancellation rates regularly.
- Use data-driven promotional campaigns.
- Maintain adequate inventory during peak demand periods.

7. Conclusion

This project successfully analyzed Amazon order data and provided valuable insights into revenue generation, customer purchasing behaviour, fulfilment performance, and geographic sales trends. Set and Kurta products emerged as the strongest business drivers, while Maharashtra and Bengaluru represented the most valuable markets.

The results indicate that a small number of product categories and regions contribute a significant share of revenue. By focusing on high-performing categories, reducing cancellations, improving fulfilment efficiency, and expanding successful regional strategies, businesses can increase revenue, improve customer satisfaction, and achieve sustainable growth.

The combination of Python, SQL, Excel, and Power BI enabled comprehensive analysis and the creation of actionable business recommendations that support data-driven decision-making.